Optimization TA 6

Kacper Krasowski

Exercise 6.3

Problem

Consider the set $X = \{\mathbf{x} \in \mathbb{R}^n : x_1 = 0\}$. Is X a linear space? What about $Y = \{\mathbf{x} \in \mathbb{R}^n : x_1 = 1\}$?

Exercise 6.3

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What about $Y = \{\mathbf{x} \in \mathbb{R}^n : x_1 = 1\}$?

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Consider the set $X = \{\mathbf{x} \in \mathbb{R}^n : x_1 = 0\}$. Is X a linear space?
What about $Y = \{\mathbf{x} \in \mathbb{R}^n : x_1 = 1\}$?

Is X a subspace of $\mathbb{R}^n$? Is Y?

Exercise 6.3: Definitions and Tools

Is X a subspace of $\mathbb{R}^n$? Is Y?

Definitions

Subspace: A subset S of a linear space X is a subspace if it satisfies

Additivity: $\mathbf{x} + \mathbf{y} \in S$ for every $\mathbf{x}, \mathbf{y} \in S$;
Homogeneity: $\alpha\mathbf{x} \in S$ for every $\mathbf{x} \in S$ and every $\alpha \in \mathbb{R}$.

Facts (Carter, Section 1.4.1)

$\mathbb{R}^n$ is a linear space with coordinatewise addition and scalar multiplication.

Every subspace is a linear space in its own right.

Every subspace contains the null vector $\mathbf{0}$.

Exercise 6.3: Sketch of the Proof

Is X a subspace of $\mathbb{R}^n$? Is Y?

1. Check additivity and homogeneity for X.

Exercise 6.3: Sketch of the Proof

Is X a subspace of $\mathbb{R}^n$? Is Y?

1. Check additivity and homogeneity for X.
2. Conclude that X is a linear space.

Exercise 6.3: Sketch of the Proof

Is X a subspace of $\mathbb{R}^n$? Is Y?

1. Check additivity and homogeneity for X.
2. Conclude that X is a linear space.
3. Find a requirement that Y violates.

Exercise 6.4

Problem

Let X be a linear space. Is it true that $\alpha\mathbf{x} = \beta\mathbf{x} \Rightarrow \alpha = \beta$? Is it true that $(\alpha - \beta)\mathbf{x} = \alpha\mathbf{x} - \beta\mathbf{x}$?

Exercise 6.4

Problem

Let X be a linear space. Is it true that $\alpha\mathbf{x} = \beta\mathbf{x} \Rightarrow \alpha = \beta$? Is it true that $(\alpha - \beta)\mathbf{x} = \alpha\mathbf{x} - \beta\mathbf{x}$?

Exercise 6.4

Problem

Let X be a linear space. Is it true that $\alpha\mathbf{x} = \beta\mathbf{x} \Rightarrow \alpha = \beta$? Is it true that $(\alpha - \beta)\mathbf{x} = \alpha\mathbf{x} - \beta\mathbf{x}$?

Does $\alpha\mathbf{x} = \beta\mathbf{x}$ imply $\alpha = \beta$ for every $\mathbf{x} \in X$?

Does $(\alpha - \beta)\mathbf{x} = \alpha\mathbf{x} - \beta\mathbf{x}$ hold for all $\mathbf{x} \in X$ and $\alpha, \beta \in \mathbb{R}$?

Exercise 6.4: Definitions

Linear space

A set X with addition and scalar multiplication is a linear space if, for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in X$ and $\alpha, \beta \in \mathbb{R}$:

Additivity

1. $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$
2. $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$
3. there is a null element $\mathbf{0}$ with $\mathbf{x} + \mathbf{0} = \mathbf{x}$
4. for every $\mathbf{x}$ there is a unique $-\mathbf{x}$ with $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$

Homogeneity

5. $(\alpha\beta)\mathbf{x} = \alpha(\beta\mathbf{x})$
6. $1\mathbf{x} = \mathbf{x}$
7. $\alpha(\mathbf{x} + \mathbf{y}) = \alpha\mathbf{x} + \alpha\mathbf{y}$
8. $(\alpha + \beta)\mathbf{x} = \alpha\mathbf{x} + \beta\mathbf{x}$

Notation: $\mathbf{x} - \mathbf{y}$ means $\mathbf{x} + (-\mathbf{y})$.

Exercise 6.4: Sketch of the Proof — First Question

$$\boxed{\text{Does } \alpha\mathbf{x} = \beta\mathbf{x} \text{ imply } \alpha = \beta \text{ for every } \mathbf{x} \in X?}$$

1. Look for a vector $\mathbf{x}$ for which the implication fails.

Exercise 6.4: Sketch of the Proof — Second Question

$$\boxed{(\alpha - \beta)\mathbf{x} = \alpha\mathbf{x} - \beta\mathbf{x} \text{ for all } \mathbf{x} \in X \text{ and } \alpha, \beta \in \mathbb{R}}$$

1. Use rules 3 and 4 to show that $(-\beta)\mathbf{x} = -(\beta\mathbf{x})$.

Exercise 6.4: Sketch of the Proof — Second Question

$$\boxed{(\alpha - \beta)\mathbf{x} = \alpha\mathbf{x} - \beta\mathbf{x} \text{ for all } \mathbf{x} \in X \text{ and } \alpha, \beta \in \mathbb{R}}$$

1. Use rules 3 and 4 to show that $(-\beta)\mathbf{x} = -(\beta\mathbf{x})$.
2. Apply rule 8 and conclude.

Exercise 6.5

Problem

Consider the space of constant real sequences $X = \{(x^n) \in \mathbb{R}^\infty : x^n = x \in \mathbb{R}\}$ where $(x^n) + (y^n) = (x + y, x + y, \ldots)$ and $\alpha(x^n) = (\alpha x, \alpha x, \ldots)$. Is this a linear space? Show that any two members $(x^n), (y^n) \in X$ are linearly dependent.

Exercise 6.5

Problem

Consider the space of constant real sequences $X = \{(x^n) \in \mathbb{R}^\infty : x^n = x \in \mathbb{R}\}$ where $(x^n) + (y^n) = (x + y, x + y, \ldots)$ and $\alpha(x^n) = (\alpha x, \alpha x, \ldots)$. Is this a linear space? Show that any two members $(x^n), (y^n) \in X$ are linearly dependent.

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Is X a subspace of $\mathbb{R}^\infty$?

Are $(x^n), (y^n) \in X$ linearly dependent?

Exercise 6.5: Definitions and Tools

Sequences (Carter, Example 1.68)

The set $\mathbb{R}^\infty$ of all real sequences is a linear space with term-by-term addition and scalar multiplication.

Definitions

Subspace: A subset S of a linear space X is a subspace if it satisfies
Additivity: $\mathbf{x} + \mathbf{y} \in S$ for every $\mathbf{x}, \mathbf{y} \in S$;
Homogeneity: $\alpha\mathbf{x} \in S$ for every $\mathbf{x} \in S$ and every $\alpha \in \mathbb{R}$.

Linearly dependent: A set S is linearly dependent if there exist distinct $\mathbf{x}_1, \ldots, \mathbf{x}_n \in S$ and numbers $\alpha_1, \ldots, \alpha_n$, not all zero, such that

$$\alpha_1\mathbf{x}_1 + \alpha_2\mathbf{x}_2 + \cdots + \alpha_n\mathbf{x}_n = \mathbf{0}.$$

Exercise 6.5: Sketch of the Proof — Linear Space

Is X a subspace of $\mathbb{R}^\infty$?

1. Compare the operations on X with those of $\mathbb{R}^\infty$.

Exercise 6.5: Sketch of the Proof — Linear Space

Is X a subspace of $\mathbb{R}^\infty$?

1. Compare the operations on X with those of $\mathbb{R}^\infty$.
2. Check additivity and homogeneity.

Exercise 6.5: Sketch of the Proof — Linear Dependence

Are $(x^n), (y^n) \in X$ linearly dependent?

1. Take two members $(x, x, \ldots)$ and $(y, y, \ldots)$.

Exercise 6.5: Sketch of the Proof — Linear Dependence

Are $(x^n), (y^n) \in X$ linearly dependent?

1. Take two members $(x, x, \ldots)$ and $(y, y, \ldots)$.
2. Find α, β, not both zero, such that $\alpha(x^n) + \beta(y^n) = \mathbf{0}$.

Exercise 6.5: Sketch of the Proof — Linear Dependence

Are $(x^n), (y^n) \in X$ linearly dependent?

1. Take two members $(x, x, \ldots)$ and $(y, y, \ldots)$.
2. Find α, β, not both zero, such that $\alpha(x^n) + \beta(y^n) = \mathbf{0}$.
3. Check the case $x = y = 0$ separately.

Exercise 6.6

Problem

Show that $\{(x_1, x_2) \in \mathbb{R}^2 : x_1 + x_2 = 1\}$ is not a linear subspace of $\mathbb{R}^2$.

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Show that $\{(x_1, x_2) \in \mathbb{R}^2 : x_1 + x_2 = 1\}$ is not a linear subspace of $\mathbb{R}^2$.

$$\boxed{S = \{(x_1, x_2) : x_1 + x_2 = 1\} \text{ is not a subspace of } \mathbb{R}^2}$$

Exercise 6.6: Sketch of the Proof

$$\boxed{S = \{(x_1, x_2) : x_1 + x_2 = 1\} \text{ is not a subspace of } \mathbb{R}^2}$$

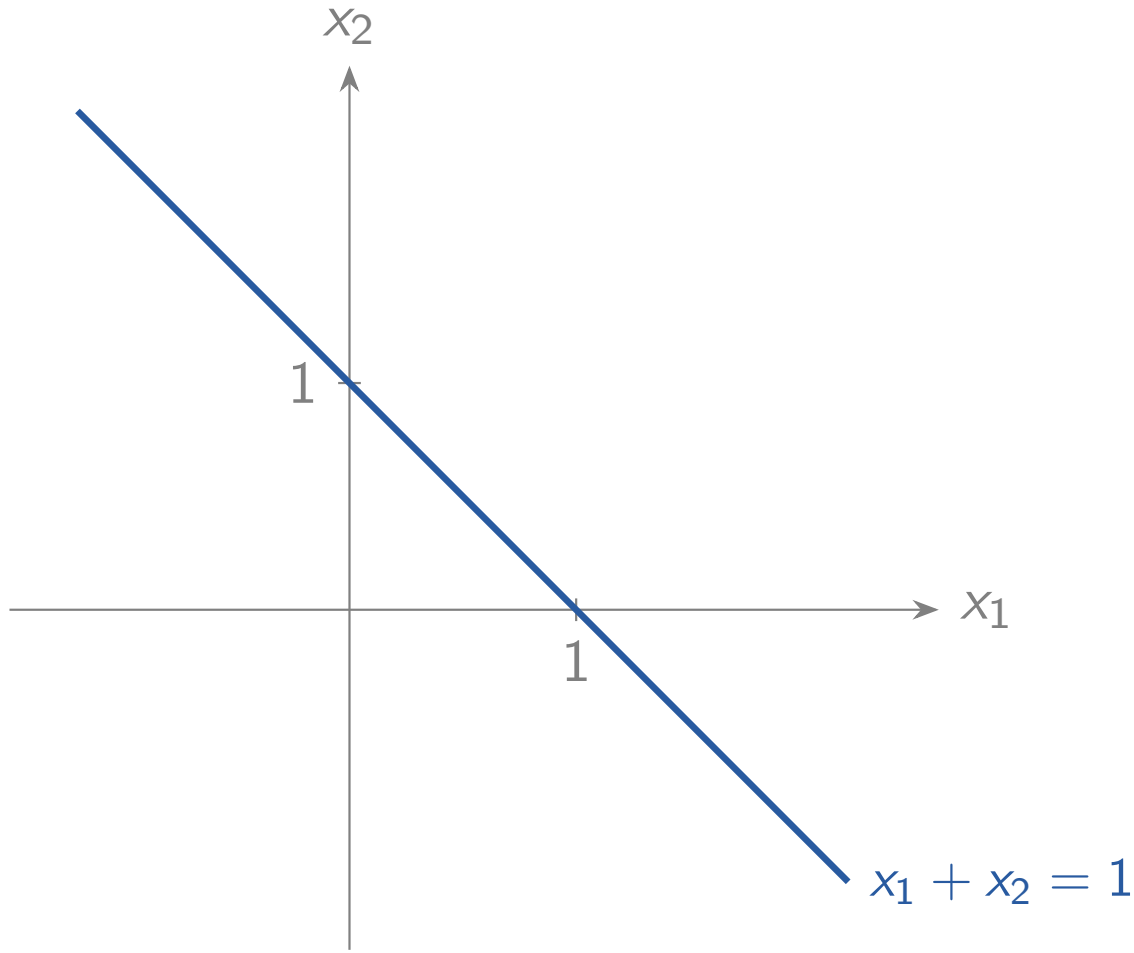
1. Check whether additivity or homogeneity fails.

Exercise 6.6: Sketch of the Proof

$$\boxed{S = \{(x_1, x_2) : x_1 + x_2 = 1\} \text{ is not a subspace of } \mathbb{R}^2}$$

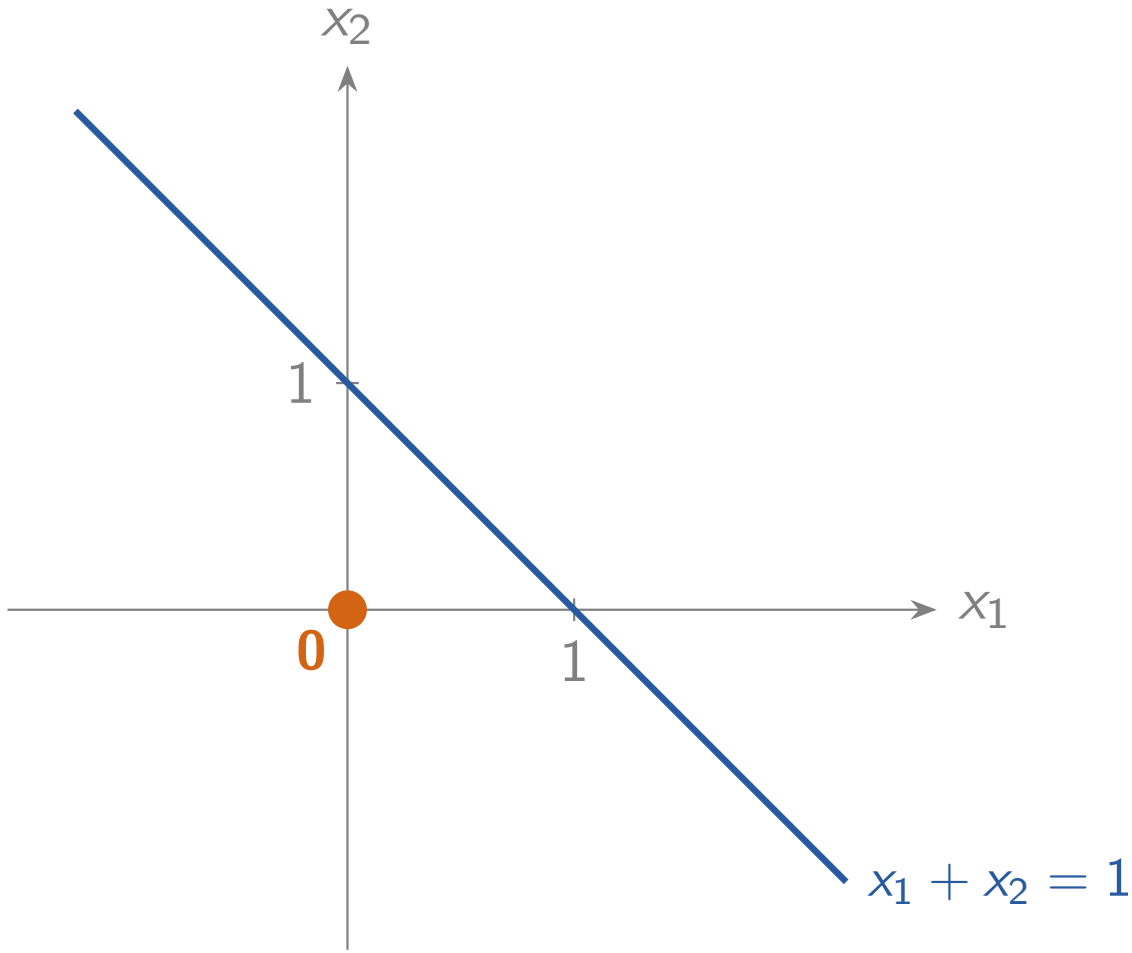
1. Check whether additivity or homogeneity fails.
2. Alternatively, use the fact that S does not contain the null vector $\mathbf{0}$.

Exercise 6.6: Intuition



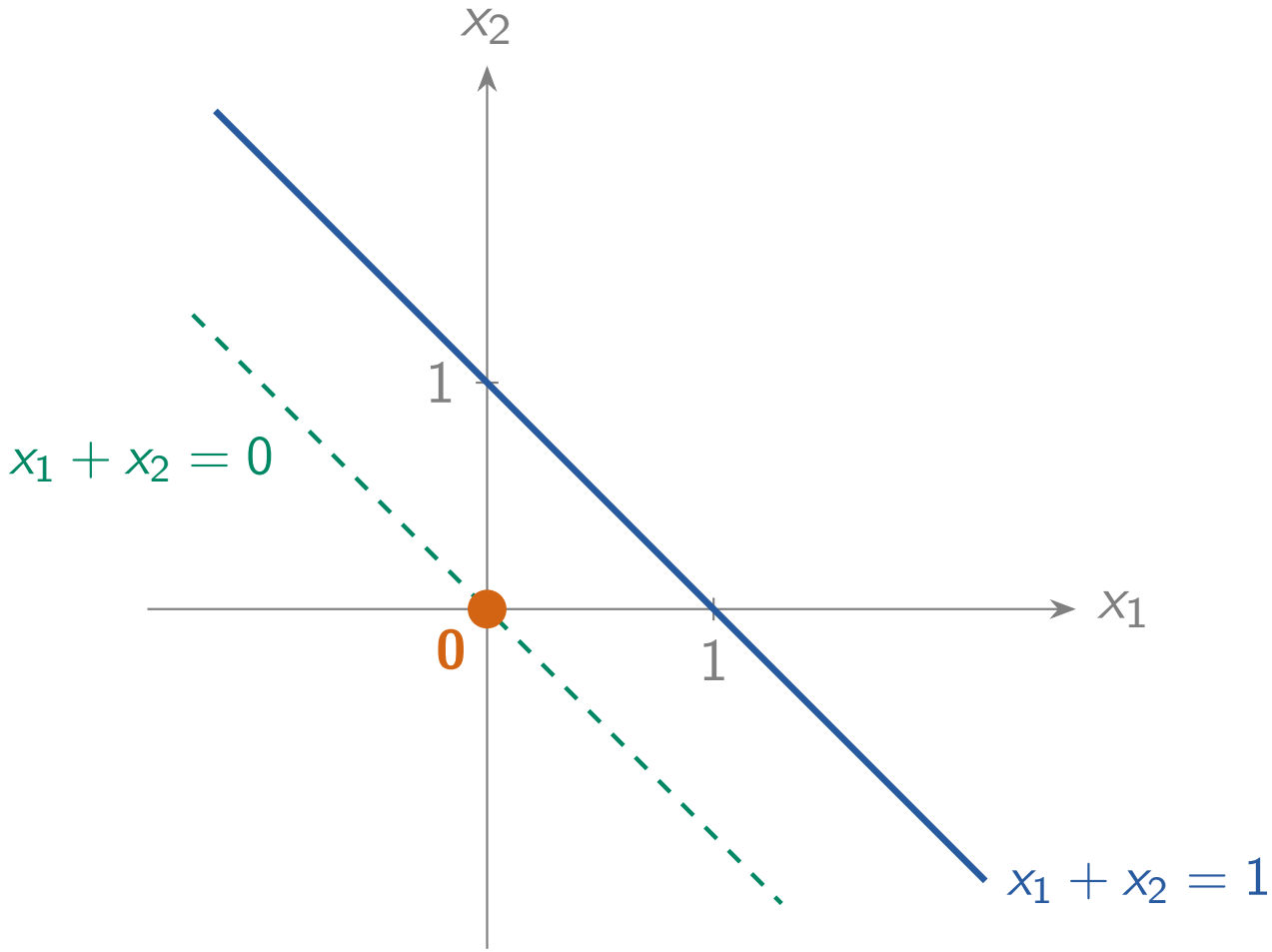
The set S is a straight line in $\mathbb{R}^2$.

Exercise 6.6: Intuition



Does the line pass through the origin?

Exercise 6.6: Intuition



S is a shifted copy of the line $x_1 + x_2 = 0$ through the origin.

Exercise 6.7

Problem

Given vectors $\mathbf{x} = (0, 1)$, $\mathbf{y} = (1, 1)$ in $\mathbb{R}^2$. Illustrate graphically the linear hull, the convex hull, an affine combination.

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Problem

Given vectors $\mathbf{x} = (0, 1)$, $\mathbf{y} = (1, 1)$ in $\mathbb{R}^2$. Illustrate graphically the linear hull, the convex hull, an affine combination.

Draw $\text{lin}\{\mathbf{x}, \mathbf{y}\}$, $\text{conv}\{\mathbf{x}, \mathbf{y}\}$, and one affine combination of $\mathbf{x}, \mathbf{y}$

Exercise 6.7: Definitions

Draw $\operatorname{lin}\{\mathbf{x}, \mathbf{y}\}$, $\operatorname{conv}\{\mathbf{x}, \mathbf{y}\}$, and one affine combination of $\mathbf{x}, \mathbf{y}$

Definitions

For $\mathbf{x}_1, \ldots, \mathbf{x}_n \in S$ and numbers $\alpha_1, \ldots, \alpha_n$, the sum $\alpha_1 \mathbf{x}_1 + \cdots + \alpha_n \mathbf{x}_n$ is a

Linear combination: for any $\alpha_i \in \mathbb{R}$;
Affine combination: if $\alpha_1 + \cdots + \alpha_n = 1$;
Convex combination: if $\alpha_i \geq 0$ and $\alpha_1 + \cdots + \alpha_n = 1$.

Hulls: The linear hull $\operatorname{lin} S$, affine hull $\operatorname{aff} S$, and convex hull $\operatorname{conv} S$ are the sets of all linear, affine, and convex combinations of elements of S, respectively.

Exercise 7.1

Problem

Let $S \subseteq X$ and $T \subseteq X$ be two bases of the same linear space X. Show that if S contains n vectors, then so must T. (Carter 1.140).

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Let $S \subseteq X$ and $T \subseteq X$ be two bases of the same linear space X. Show that if S contains n vectors, then so must T. (Carter 1.140).

$$\boxed{S, T \text{ bases of } X, \ S \text{ has } n \text{ vectors} \implies T \text{ has } n \text{ vectors}}$$

Exercise 7.1: Definitions

$$\boxed{S, T \text{ bases of } X,\ S \text{ has } n \text{ vectors} \implies T \text{ has } n \text{ vectors}}$$

Definitions

Linear span: The linear span (linear hull) of $S \subseteq X$ is the set of all linear combinations of elements of S,

$$\operatorname{lin} S = \{\alpha_1 \mathbf{x}_1 + \cdots + \alpha_m \mathbf{x}_m : \mathbf{x}_1, \ldots, \mathbf{x}_m \in S,\ \alpha_1, \ldots, \alpha_m \in \mathbb{R}\}.$$

Linearly independent: A set $S \subseteq X$ is linearly independent if, for all distinct $\mathbf{x}_1, \ldots, \mathbf{x}_m \in S$ and all numbers $\alpha_1, \ldots, \alpha_m$,

$$\alpha_1 \mathbf{x}_1 + \cdots + \alpha_m \mathbf{x}_m = \mathbf{0} \implies \alpha_1 = \cdots = \alpha_m = 0.$$

Basis: A set $S \subseteq X$ is a basis for X if it is linearly independent and spans X, that is, $\operatorname{lin} S = X$.

Exercise 7.1: Sketch of the Proof

$$\boxed{S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\},\ T = \{\mathbf{w}_1, \ldots, \mathbf{w}_m\} \text{ bases of } X \implies m = n}$$

1. Let $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ and $T = \{\mathbf{w}_1, \ldots, \mathbf{w}_m\}$.

Exercise 7.1: Sketch of the Proof

$$\boxed{S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\},\ T = \{\mathbf{w}_1, \ldots, \mathbf{w}_m\} \text{ bases of } X \implies m = n}$$

1. Let $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ and $T = \{\mathbf{w}_1, \ldots, \mathbf{w}_m\}$.
2. Show that $\mathbf{w}_1$ is a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_n$.

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2. Show that $\mathbf{w}_1$ is a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_n$.
3. Rearrange and show that some $\mathbf{v}_i$ can be substituted by $\mathbf{w}_1$, and the new set is still a basis.

Exercise 7.1: Sketch of the Proof

$$\boxed{S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\},\ T = \{\mathbf{w}_1, \ldots, \mathbf{w}_m\} \text{ bases of } X \implies m = n}$$

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4. Repeat n times.

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3. Rearrange and show that some $\mathbf{v}_i$ can be substituted by $\mathbf{w}_1$, and the new set is still a basis.
4. Repeat n times.
5. Show that the new basis has n elements, all from T.

Exercise 7.1: Sketch of the Proof

$$\boxed{S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\},\ T = \{\mathbf{w}_1, \ldots, \mathbf{w}_m\} \text{ bases of } X \implies m = n}$$

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3. Rearrange and show that some $\mathbf{v}_i$ can be substituted by $\mathbf{w}_1$, and the new set is still a basis.
4. Repeat n times.
5. Show that the new basis has n elements, all from T.
6. Argue why it must be that $m \leq n$.

Exercise 7.1: Sketch of the Proof

$$S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\},\ T = \{\mathbf{w}_1, \ldots, \mathbf{w}_m\} \text{ bases of } X \implies m = n$$

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3. Rearrange and show that some $\mathbf{v}_i$ can be substituted by $\mathbf{w}_1$, and the new set is still a basis.
4. Repeat n times.
5. Show that the new basis has n elements, all from T.
6. Argue why it must be that $m \leq n$.
7. Repeat steps 2–6 with the roles of S and T swapped to show that $n \leq m$.

Exercise 7.1: Sketch of the Proof

$$S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\},\ T = \{\mathbf{w}_1, \ldots, \mathbf{w}_m\} \text{ bases of } X \implies m = n$$

1. Let $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ and $T = \{\mathbf{w}_1, \ldots, \mathbf{w}_m\}$.
2. Show that $\mathbf{w}_1$ is a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_n$.
3. Rearrange and show that some $\mathbf{v}_i$ can be substituted by $\mathbf{w}_1$, and the new set is still a basis.
4. Repeat n times.
5. Show that the new basis has n elements, all from T.
6. Argue why it must be that $m \leq n$.
7. Repeat steps 2–6 with the roles of S and T swapped to show that $n \leq m$.
8. Conclude that $m = n$.

Exercise 7.2

Problem

We define the conic hull of a subset $S \subseteq X$ by the collection of nonnegative linear combinations

$$\text{cone}(S) = \left\{ \sum_{i=1}^{n} \alpha_i \mathbf{x}_i : \mathbf{x}_i \in S,\ \alpha_i \geq 0 \right\}.$$

What is the conic hull of the vectors $(0,1), (1,0)$ in $\mathbb{R}^2$?

Exercise 7.2

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We define the conic hull of a subset $S \subseteq X$ by the collection of nonnegative linear combinations

$$\text{cone}(S) = \left\{ \sum_{i=1}^{n} \alpha_i \mathbf{x}_i : \mathbf{x}_i \in S,\ \alpha_i \geq 0 \right\}.$$

What is the conic hull of the vectors $(0,1), (1,0)$ in $\mathbb{R}^2$?

Describe $\text{cone}(S)$ for $S = \{(0,1),(1,0)\} \subseteq \mathbb{R}^2$

Exercise 7.3a

Problem

Consider the set of affine functionals
$X = \{f : \mathbb{R} \to \mathbb{R} \text{ with } f(x) = ax + b \text{ and } a, b \in \mathbb{R}\}$.

(a) Is X a linear space?

Exercise 7.3a: Definitions and Tools

Function space

The set of all functions $f : \mathbb{R} \to \mathbb{R}$ is a linear space with

$$(f + g)(x) = f(x) + g(x), \qquad (\alpha f)(x) = \alpha f(x).$$

Its null vector is the zero function.

Definitions

Subspace: A subset S of a linear space X is a subspace if it satisfies
Additivity: $\mathbf{x} + \mathbf{y} \in S$ for every $\mathbf{x}, \mathbf{y} \in S$;
Homogeneity: $\alpha\mathbf{x} \in S$ for every $\mathbf{x} \in S$ and every $\alpha \in \mathbb{R}$.

Exercise 7.3a: Sketch of the Proof

1. View X as a subset of the linear space of all functions $f : \mathbb{R} \to \mathbb{R}$.

Exercise 7.3a: Sketch of the Proof

1. View X as a subset of the linear space of all functions $f : \mathbb{R} \to \mathbb{R}$.
2. Check additivity and homogeneity.

Exercise 7.3b

Problem

Consider the set of affine functionals
$X = \{f : \mathbb{R} \to \mathbb{R} \text{ with } f(x) = ax + b \text{ and } a, b \in \mathbb{R}\}$.

(b) Provide an example of a linear subspace of X.

Exercise 7.3b: Sketch of the Proof

1. Fix a or b (or both) and see what works.

Exercise 7.3b: Sketch of the Proof

1. Fix a or b (or both) and see what works.
2. Check additivity and homogeneity.

Exercise 7.3c

Problem

Consider the set of affine functionals
$X = \{f : \mathbb{R} \to \mathbb{R} \text{ with } f(x) = ax + b \text{ and } a, b \in \mathbb{R}\}$.

(c) Find a basis of X and $\dim(X)$.

Exercise 7.3c: Definitions

Definitions

Basis: A set $B \subseteq X$ is a basis for X if it is linearly independent and spans X, that is, $\operatorname{lin} B = X$ (Exercise 7.1).

Dimension: If X has a basis with n elements, then $\dim X = n$. By Exercise 7.1, this does not depend on the basis.

Exercise 7.3c: Sketch of the Proof

1. Propose two functions that span X.

Exercise 7.3c: Sketch of the Proof

1. Propose two functions that span X.
2. Show that they are linearly independent.

Exercise 7.3c: Sketch of the Proof

1. Propose two functions that span X.
2. Show that they are linearly independent.
3. Conclude $\dim X$.

Exercise 7.3d

Problem

Consider the set of affine functionals
$X = \{f : \mathbb{R} \to \mathbb{R} \text{ with } f(x) = ax + b \text{ and } a, b \in \mathbb{R}\}$.

(d) Consider $Y = \{f : \mathbb{R} \to \mathbb{R} \text{ with } f(x) = ax + 1 \text{ and } a \in \mathbb{R}\}$ and how it relates to X. Is it a subspace, affine space, hyperplane in X?

Exercise 7.3d: Definitions

Definitions

Affine set: A subset S of a linear space X is affine if, for every $\mathbf{x}, \mathbf{y} \in S$ and every $\alpha \in \mathbb{R}$,

$$\alpha\mathbf{x} + (1 - \alpha)\mathbf{y} \in S.$$

For every affine set S, there is a unique subspace V such that $S = \mathbf{x}_0 + V$ for some $\mathbf{x}_0 \in S$. We say S is parallel to V, and the dimension of S is $\dim V$.

Hyperplane: A maximal proper affine subset of X. If $\dim X = n$, the hyperplanes are exactly the affine sets of dimension $n - 1$.

Exercise 7.3d: Sketch of the Proof

1. Check whether Y is a subspace of X.

Exercise 7.3d: Sketch of the Proof

1. Check whether Y is a subspace of X.
2. Check whether Y is an affine set.

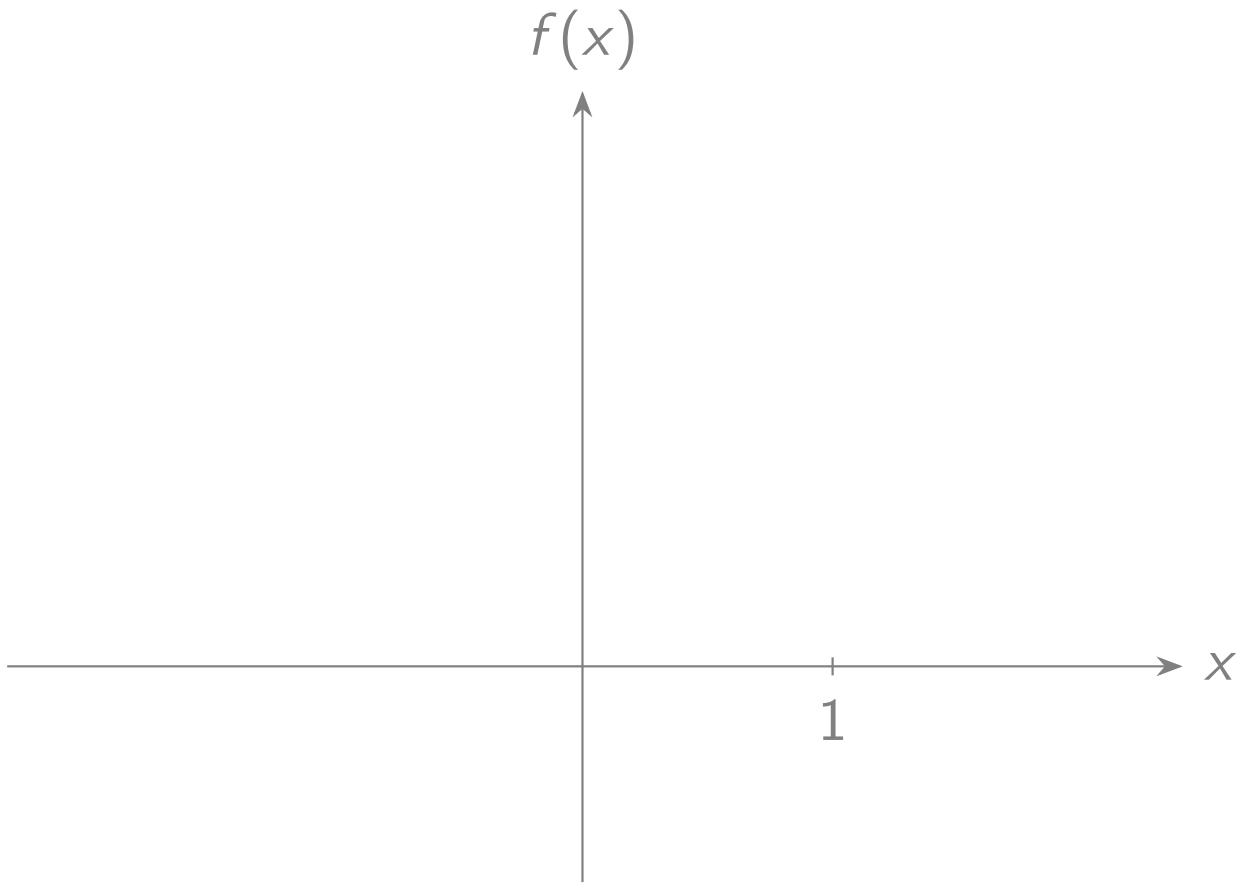
Exercise 7.3d: Sketch of the Proof

1. Check whether Y is a subspace of X.
2. Check whether Y is an affine set.
3. Find the subspace V parallel to Y and show that $\dim V = 1$.

Exercise 7.3d: Sketch of the Proof

1. Check whether Y is a subspace of X.
2. Check whether Y is an affine set.
3. Find the subspace V parallel to Y and show that $\dim V = 1$.
4. Conclude that $\dim Y = \dim X - 1$, so Y is a hyperplane in X.

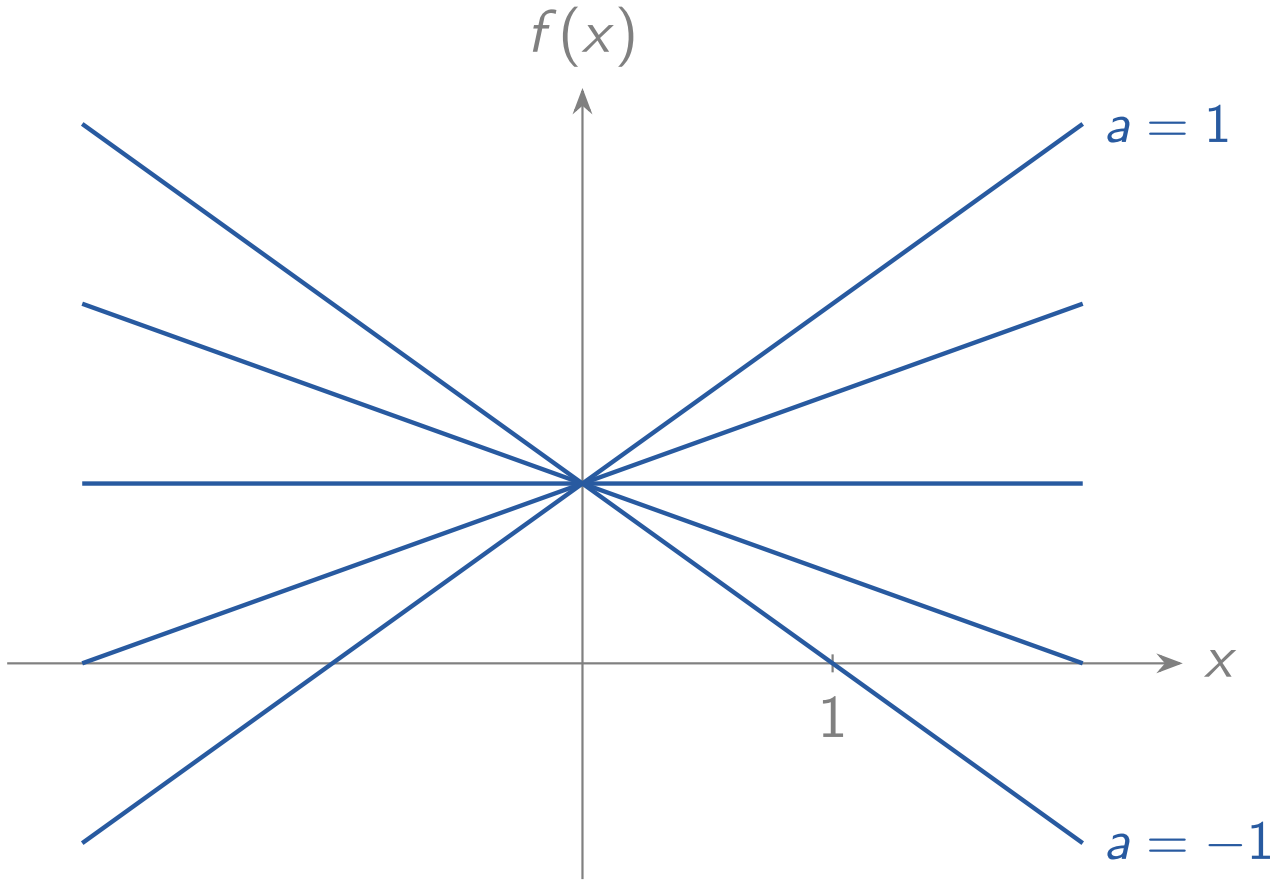
Exercise 7.3d: Intuition



Each element of X is a function, drawn as its graph.

Exercise 7.3d: Intuition

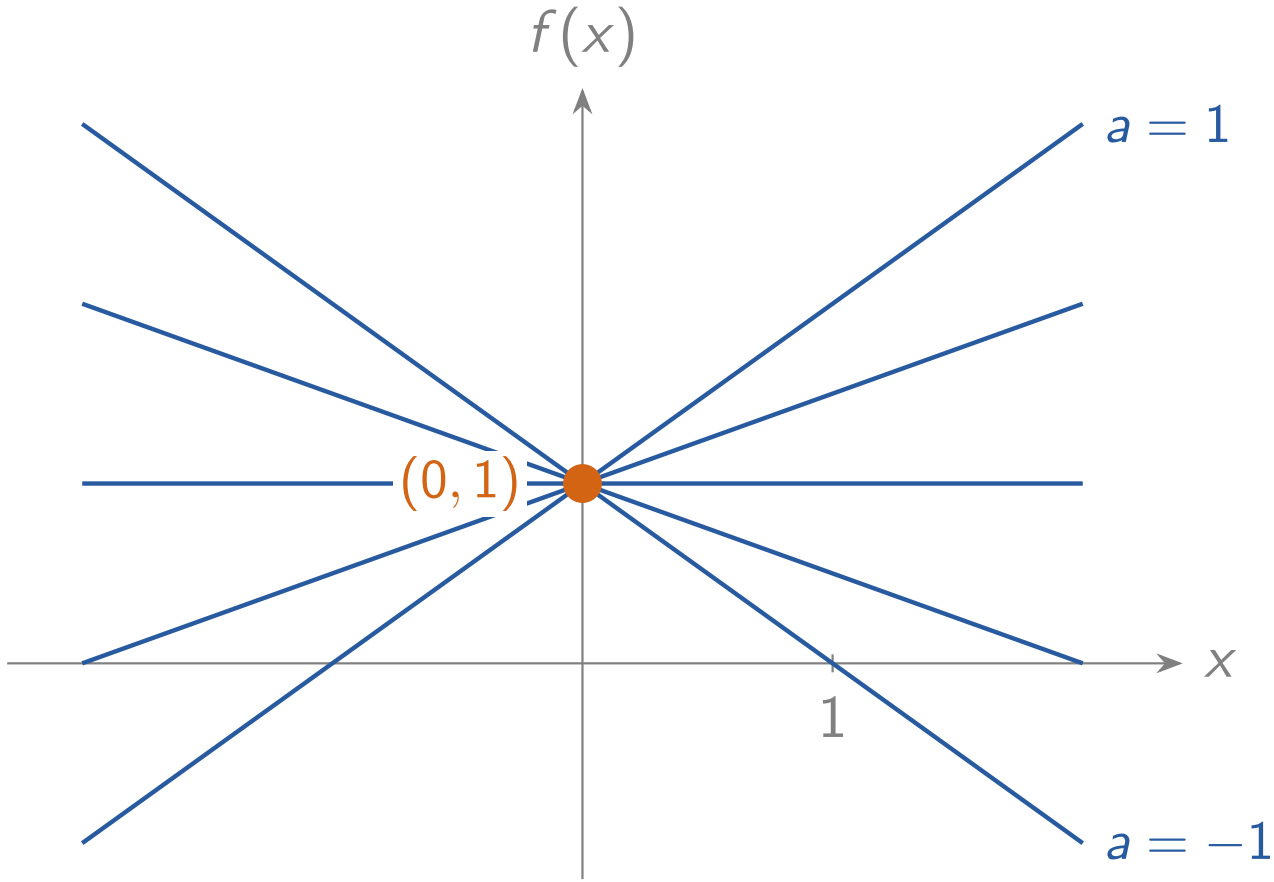
Members of Y: $f(x) = ax + 1$ for $a \in \{-1, -\frac{1}{2}, 0, \frac{1}{2}, 1\}$.



The graphs of some members of Y.

Exercise 7.3d: Intuition

Members of Y: $f(x) = ax + 1$ for $a \in \{-1, -\frac{1}{2}, 0, \frac{1}{2}, 1\}$.



Every member of Y passes through $(0, 1)$. What about the zero function?

Exercise 7.4a

Problem

Consider a linear function $f : \mathbb{R}^n \to \mathbb{R}^m$.

(a) Show that it can be represented by $f(\mathbf{x}) = A\mathbf{x}$, where A is an $m \times n$ matrix. [Hint: We can write $f(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_m(\mathbf{x}))^\top$, where $f_i : \mathbb{R}^n \to \mathbb{R}$ is a linear functional that maps $\mathbf{x}$ to the i-th coordinate of $f(\mathbf{x})$. It suffices to show $f_i(\mathbf{x}) = \sum_{j=1}^{n} a_{ij} x_j$ where x_j is the j-th coordinate of $\mathbf{x}$.]

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$$f : \mathbb{R}^n \to \mathbb{R}^m \text{ linear } \implies f(\mathbf{x}) = A\mathbf{x} \text{ for some } m \times n \text{ matrix } A$$

Exercise 7.4a: Definitions and Tools

$$\boxed{f : \mathbb{R}^n \to \mathbb{R}^m \text{ linear } \implies f(\mathbf{x}) = A\mathbf{x} \text{ for some } m \times n \text{ matrix } A}$$

Definitions

Linear function: A function $f : X \to Y$ between linear spaces is linear if, for every $\mathbf{x}_1, \mathbf{x}_2 \in X$ and every $\alpha \in \mathbb{R}$,

$$f(\mathbf{x}_1 + \mathbf{x}_2) = f(\mathbf{x}_1) + f(\mathbf{x}_2), \qquad f(\alpha\mathbf{x}_1) = \alpha f(\mathbf{x}_1).$$

Linear functional: A linear function $g : X \to \mathbb{R}$.

Recall

Every linear functional $g : \mathbb{R}^n \to \mathbb{R}$ can be represented as an inner product, $g(\mathbf{x}) = \mathbf{a} \cdot \mathbf{x}$ for some $\mathbf{a} \in \mathbb{R}^n$.

Exercise 7.4a: Sketch of the Proof

$$\boxed{f : \mathbb{R}^n \to \mathbb{R}^m \text{ linear } \implies f(\mathbf{x}) = A\mathbf{x} \text{ for some } m \times n \text{ matrix } A}$$

1. Write $f(\mathbf{x}) = (f_1(\mathbf{x}), \ldots, f_m(\mathbf{x}))^\top$.

Exercise 7.4a: Sketch of the Proof

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1. Write $f(\mathbf{x}) = (f_1(\mathbf{x}), \ldots, f_m(\mathbf{x}))^\top$.
2. Each f_i is a linear functional, so express it as $f_i(\mathbf{x}) = \mathbf{a}_i \cdot \mathbf{x}$.

Exercise 7.4a: Sketch of the Proof

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3. Stack the vectors $\mathbf{a}_i^\top$ as the rows of A and conclude that $f(\mathbf{x}) = A\mathbf{x}$.

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3. Stack the vectors $\mathbf{a}_i^\top$ as the rows of A and conclude that $f(\mathbf{x}) = A\mathbf{x}$.

Note: the entries are $a_{ij} = f_i(\mathbf{e}_j)$.